Topology: real head families, weak depth order, a matching-pair loop

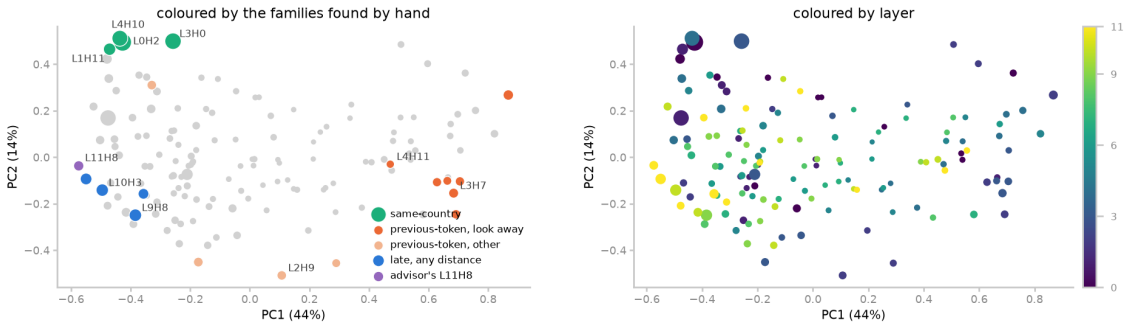
Four pre-registered topological analyses of Exp 4 / 4b (GPT-2 small, 200 random rows × 11 country–capital pairs; country at τ or not; final token T = its capital, a wrong capital, or a random token), each checked independently.

Conclusion

- **Head types are real families, not tight clusters.** Persistent homology and Mapper chain points together and see distinctive heads as isolated individuals. Clustering that does not chain recovers the families found by hand. The same-country heads always group (pair rate 1.00 vs 0.13–0.21 by chance). The previous-token heads split into “look away from a matching neighbour” and “read any nearby word”.
- **There are no loops in behaviour space, and depth orders behaviour only weakly.** No continuous families exist. Communities track layer only weakly (normalised mutual information 0.11–0.28), so “layers 2–3 local, 9–11 global” rests on a few heads.
- **Curve shapes are measured, not eyeballed.** All 5 checks pass against a calibrated null. L11H8 has one decaying peak. L3H0 *rises* with distance (0.14 → 0.41 / 0.51). L9H8 spikes at both ends. No previous-token head has a feature beyond 8 tokens.
- **Only the matching capital closes a loop through its country in the attention graph.** This happens in layer 0 at τ = 16, 8 and 1 and in layer 4 at τ = 24: in every bootstrap with its capital, never with a wrong one. In layers 1, 3 and 4 loops also form with a wrong capital, meaning only “a country is present”. With the country adjacent, no loop forms; only the weights change.

1 · Families sit together; layers mix

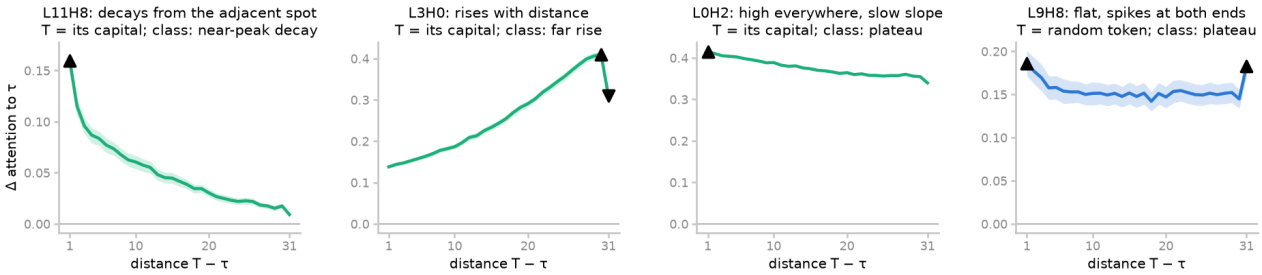
each dot = one head, size = response strength



Heads placed by their whole response (all difference curves, shape-normalised; display only, tests ran in full space).

2 · Shapes of the distance curves

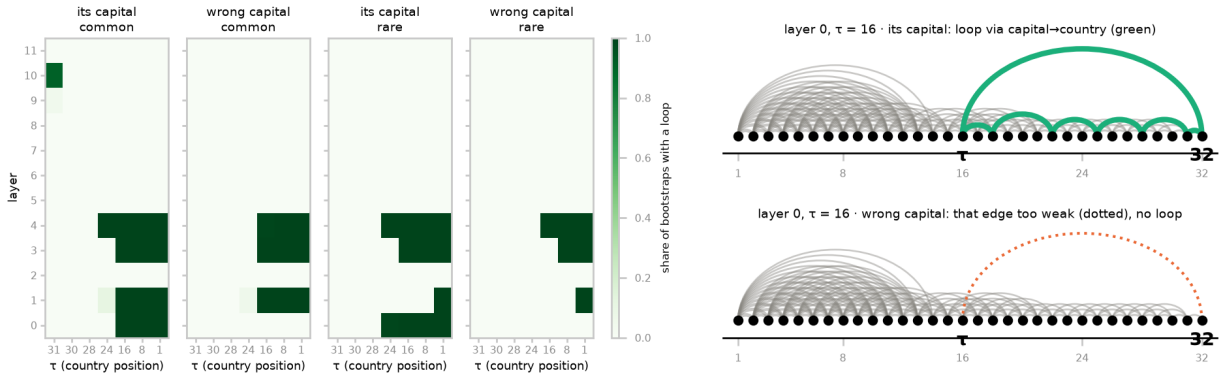
triangles = significant peaks / dips (row-bootstrap null, 99th pct, ≥ 0.02)



L0H2 is a “plateau” only by the relative rule: it declines steadily, 0.42 → 0.34.

3 · Loops through the capital→country edge

BOS removed; edges shown above the weight where the loop is born



Left: share of bootstraps with a persistent loop through that edge, by layer and country position. Right: in layer 0 the capital’s edge to its own country completes a loop with the chain of local links.

Verdicts: head types T1-a/b/c FAIL, T1-d PASS; Mapper all FAIL or unstable (a method artifact); curves C1–C5 PASS (C4 depends on the definition); graph G1, G2, G3b, B1 PASS, G3a FAIL.

Caveats: the same 11 pairs throughout (consistent ≠ large); graphs are averages over sequences. Details: exp4/LOG.md.